ClothPit

USER DOCUMENTATION

Cloth mesh generation, pinning, simulation controls and wind



DOCUMENT DETAILS	
Version	0.2.6
Blender compatibility	Blender 5.2 (declared in add-on metadata)
Author	pit_

Installation, feature reference, workflows and troubleshooting.

1. Installation

The add-on is distributed as a single Python file. Install it from Blender Preferences using the add-on installation workflow available in your Blender version.

- Open Blender 5.2.
- Go to Edit → Preferences.
- Open the add-on installation interface (Get Extensions / Install from Disk, depending on the Blender build).
- Select **cloth_pit_v0_2_6(2).py**.
- Enable the add-on after installation if Blender does not enable it automatically.

Interface location: 3D Viewport → Sidebar (N) → ClothPit

If the add-on does not appear, check the Blender version, confirm that the file is the actual .py add-on, and inspect the system console for registration errors.

2. Generate Cloth

Generate Cloth creates a subdivided mesh intended for cloth simulation. The panel exposes mesh generation settings and fabric presets.

- Set the mesh generation parameters in section 1. Generate Cloth.
- Choose Generate Static Cloth Mesh to create the mesh without relying on an existing cloth object.
- Use Apply Preset to apply a starting cloth profile to the selected mesh.

Available fabric profiles in the current source include Fabric, Cotton, Leather and Canvas. Presets are practical starting values, not physically measured material definitions.

The add-on provides Fabric Type selection; use the presets exposed by the installed build.

3. Pinning Tools

Pinning tools use vertex groups to hold selected parts of a cloth mesh in place during simulation.

- Select the cloth mesh and enter Edit Mode.
- Select the vertices to pin.
- Use Pin Selected to assign the selected vertices to the Cloth_Pin vertex group.
- Use Select Pinned to retrieve the pinned selection.
- Use Unpin Selected or Clear Pinning to remove pin assignments.

Pin and Unpin operations require an active mesh and a vertex selection in Edit Mode.

4. Collision Tools

Collision Tools configure collision objects and self-collision settings for the cloth simulation.

- Set collision distance and friction, then use Generate Collision to configure a selected collision object.
- Set self-collision distance and friction, then choose Enable Self Collision on the cloth object.
- Use Clear Collision to remove the collision setup created by the tool.

Self-collision availability and behavior depend on the Blender build and the cloth physics API exposed by that build.

5. Modifiers

- **Solidify:** set a thickness value and add a Solidify modifier to give the cloth surface thickness.
- **Subdivision:** set viewport and render subdivision levels, then add a Subdivision Surface modifier.

Modifier order affects the result. Review the modifier stack after adding thickness or subdivision, especially when the mesh is also simulated.

6. Simulation Controls

Control	Purpose
Start / End	Set the scene timeline range used for the simulation.
Play / Pause	Start or pause timeline playback.
Reset	Return the timeline to its start frame.
Freeze	Keep the geometry at the current frame by storing the current evaluated pose and disabling the live cloth modifier. It does not reset the simulation.
Unfreeze	Restore the live cloth simulation state from the frozen pose.
Bake / Apply	Apply the Cloth modifier at the current frame, converting the current evaluated result into mesh geometry.
Clear Bake	Clear the baked simulation cache.

Freeze and Bake / Apply are different operations: Freeze preserves a temporary pose while keeping the simulation workflow reversible; Bake / Apply applies the modifier at the current frame.

7. Wind Generator

Wind Generator creates a Blender Wind force field. Set Strength, Flow and Noise before choosing Create Wind. When a Wind object is selected, the controls update that selected field; otherwise, they define the values for a new Wind object.

8. Troubleshooting and Notes

Situation	Suggested check
Pinning does nothing	Confirm that a mesh is active and selected vertices are available in Edit Mode.
Cloth does not move	Check that the object has a Cloth modifier, timeline range is valid, and playback starts from the intended frame.
Collision is ineffective	Check collision object setup, distances, friction and object scale.
Self Collision unavailable	Some Blender builds or API configurations may not expose the required setting. The add-on can only enable it when supported.
Wind has no visible effect	Check the selected Wind field, its orientation, strength, flow and noise, as well as the cloth simulation settings.
Unexpected mesh result	Review mesh density, modifier order, pin group membership and the current frame before applying the simulation.

ClothPit · pit_ · Version 0.2.6 · Blender 5.2